

DOES A HAMILTONIAN FILE EXIST? YES: 1; NO: 0

- for numerical efficiency; if you're just repeating a previous calculation with different values for mass m or coupling g , this allows you to reuse the previous calculation; can select 0 on a first pass

INPUT N:

- number of colors (can be anything; 3 is for qcd)

INPUT NF:

- number of flavors (diff quarks); each mass is set separately later

INPUT B:

- baryon number for states

INPUT LAMBDA ($0.0 < \text{LAMBDA} < 1.0$):

- lambda is the dimensionless coupling based on the ratio of the usual coupling g (which carries mass dimension in 1+1) and the first quark mass. It runs from 1 (infinite coupling) to 0 (zero coupling)
 $\text{lambda} = \sqrt{1/(1 + \pi^2 m^2/g^2)}$ (with m the first quark mass)

input ratio of quark mass 2 to quark mass 1:

- give other masses relative to the first one

input flavor qm number for flavor 1

- flavor number of type one for states (can be positive or negative)

CUTOFF/MASS SQD: (LE.0 = NO CUTOFF

- can introduce a cutoff in energy to limit states; safe to leave at no cutoff

LIMIT PARTICLE NUMBER TO: (0 = NO LIMIT

- the number of states can grow rapidly; this allows you to limit by hand the number of particles in higher-fock states. For mesons, you might start with 2 or 4, then allow for more later until it gets out of hand.

INPUT K (GE 2):

- K is twice the total numerical lc momentum k^+ of the state. It also serves as a discretization parameter; the larger K, the closer to the continuum.

The program should run at this point and give you updates as various parts are computed. The output will include a listing of the states in your space, and eigenvalues and eigenvectors, among other bits of information.

The states are restricted to color singlets. The notation in the four rows is for four particles in this case. The top two rows have 1 in the top row for a quark, one in the second row for an antiquark. The third row gives 2 k^+ carried by that quark (k^+ is half-integer due to antiper. bcs). The last row gives the flavor.

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STATE NUMBER:      5
  LGTH,NMES,NBRB,NBRD,KMES,KBRB,KBRD
    4    2    0    0    8    0    0
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    1    0    1    0
    0    1    0    1
    1    1    1    5
    1    1    1    2
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Here the quark-antiquark pairs are contracted into color singlets; states are always singlets.

It turns out states generated in the obvious way were over-complete and not orthonormal, so the first thing the code does is weed out redundant states and figure out the orthonormal combinations of those that are retained. That information is contained in the matrix Z , also in the output.